

Pharmacology of Adrenergic Antagonists:

- **Alpha-adrenergic antagonists (Alpha blockers)**
- **Beta-adrenergic antagonists (Beta blockers)**

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completion of preparation materials, students should be able to:

1. Subclasses

Identify subclasses of the individual adrenergic blockers based on their receptor selectivity.

2. Pharmacokinetic properties → Patient care

List to ADME properties of starred drugs and their relevance to patient care.

3. Epinephrine effect → Clinical rationale

Explain the effects and clinical relevance of epinephrine reversal and compare to those of phenylephrine.

4. Mechanisms of action → Cardiovascular effects

Characterize the cardiovascular effects and reflex responses with administration of each of the following as monotherapy by giving examples of each: a nonselective alpha blocker, a selective alpha-1 blocker, a selective alpha-2 agonist, a nonselective beta-blocker, a selective beta-1 blocker, and a beta blocker with combined β and α blocking actions.

5. Mechanisms of action → Effects → Therapeutic uses

Correlate the mechanisms of action and pharmacological effects of the adrenergic receptor antagonists to the agents' specific therapeutic uses.

6. Mechanisms → Adverse effects → Cautious use

Relate the mechanisms of adverse effects and potential drug interactions of the individual agents to their cautions and contraindications for use.

Preparation Materials

Required

ScholarRx Bricks | Practice Questions

Suggested supplemental resources if you wish to use them:

Video lecture | Dr. Goldstein's Word handout | Guided Reading Questions

Links to textbook and pharmacology review books are provided on the next slide.

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them. (ScholarRx Bricks & Practice Questions are required.)*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links

ScholarRx Brick

Brick: **General Pharmacology > Autonomic Drugs > Drugs that Alter Sympathetic Tone**

- Sympatholytics > α -Adrenergic Agents: <https://exchange.scholarrx.com/brick/945-adrenergic-agents>
- Sympatholytics > β -Blockers <https://exchange.scholarrx.com/brick/946-blockers> Access Medicine

Katzung's Basic & Clinical Pharmacology, 16e, 2022: Chapter 10: Adrenoceptor Antagonist Drugs

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit/content.aspx?bookid=3382§ionid=281747971>

FYI: Review books provide practice questions.

Access Medicine Katzung's Pharmacology: Examination & Board Review, 13e, 2021; Chapter 10:

Adrenoceptor Blockers

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285590454>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 7: Adrenergic Antagonists

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit/content.aspx?sectionid=253325856&bookid=3222>

What you need to know and understand about adrenergic blockers

- The alpha blockers relax arterial and venous smooth muscle and decrease vascular resistance, whereas the beta blockers reduce heart rate and cardiac output.
- Both alpha and beta blockers reduce blood pressure.
- The nonselective alpha blockers are phenoxybenzamine (an irreversible inhibitor) and phentolamine (a competitive blocker). These drugs block both α_1 - and α_2 -adrenergic receptors.
- Phenoxybenzamine is primarily used to manage hypertensive episodes caused by pheochromocytoma. Phentolamine is for intraoperative severe hypertension, reversal of extravasation by vasopressors, and reversal of oral soft tissue local anesthesia.
- The nonselective alpha blockers cause marked tachycardia due to the reflex response to vasodilation as well as causing NE release by inhibition of presynaptic α_2 receptors, which activates cardiac beta-1 receptors, and they cause marked orthostatic hypotension. They also can cause miosis, nasal stuffiness, and impaired/retrograde ejaculation.

What you need to know and understand about adrenergic blockers

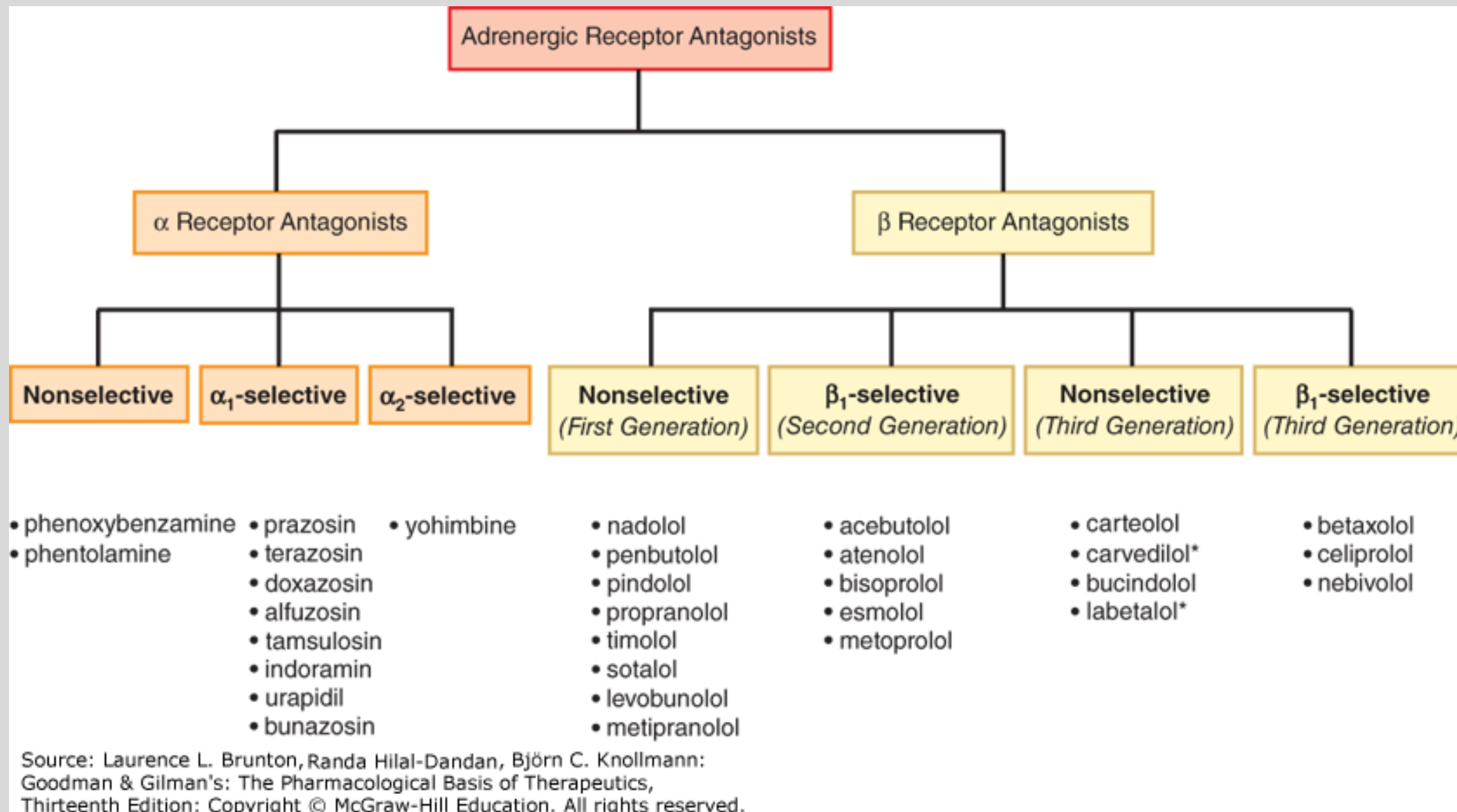
- The selective alpha-1 blockers prazosin, terazosin, and doxazosin (quinazolines) are adjunctive (add-on) agents in the management of chronic primary hypertension or urinary obstruction caused by benign prostatic hyperplasia (BPH) and other conditions.
- The major adverse effect of these drugs is marked orthostatic hypotension with the first dose (first-dose effect). Relaxation of iris dilator muscle due to α_1 antagonism can cause intraoperative floppy iris syndrome. The drugs can also cause nasal stuffiness due to alpha blockade in the vasculature of the nasal mucosa, and impaired ejaculation.
- Tamsulosin and silodosin are selective α_{1A} receptor inhibitors used for the management of urinary symptoms of BPH. Alfuzosin is a quinazoline α_{1A} receptor inhibitor for symptomatic treatment of BPH. These drugs are not used for hypertension.
- Yohimbine is an alkaloid derived from a botanical source. It is an α_2 receptor competitive antagonist with effects opposite those of the α_2 receptor agonist clonidine. It is available in herbal products for erectile dysfunction (never clearly demonstrated in clinical trials).

What you need to know and understand about adrenergic blockers

- Beta blockers antagonize the effects of catecholamines at β -adrenergic receptors. Evidence suggests that some agents are inverse agonists.
- The beta blockers differ in their relative affinities for β ARs. The nonselective beta blockers inhibit both β_1 and β_2 receptors, whereas the selective agents inhibit β_1 ARs at usual therapeutic doses. None of the clinically available β_1 -selective agents is completely selective. Selectivity tends to diminish at higher drug concentrations.
- Some actions of beta blockers may be due to additional effects, such as partial agonist action, α_1 AR blockade, and membrane stabilizing effect.
- The beta blockers are well absorbed from the GI tract and undergo extensive hepatic metabolism. First-pass effect varies among the individual agents. They are rapidly and widely distributed to the tissues. Lipophilic beta blockers, propranolol, metoprolol, and nebivolol, distribute to the CNS and may cause CNS effects, such as fatigue, depression, and insomnia. CNS effects may contribute to blood pressure lowering effects.

What you need to know and understand about adrenergic blockers

- Beta blockers have prominent effects on the heart, decreasing heart rate (chronotropy) and force of contraction (inotropy), and reducing the rate of conduction through the AV node. They inhibit β_1 receptors on specialized cells in the kidney, which suppresses renin release. (Renin increases blood pressure by increasing angiotensin II and aldosterone – the renin, angiotensin, aldosterone system, RAAS.)
- Beta blockers have a variety of clinical applications, including the treatment of hypertension, angina pectoris, myocardial infarction, cardiac arrhythmias, congestive heart failure, glaucoma, and the prevention of migraine headache.
- Mechanism-related adverse effects include bradycardia, bronchoconstriction, potentiation or masking of hypoglycemia, and lipid abnormalities.
- Abrupt withdrawal of beta blocker therapy should be avoided because it can cause severe rebound hypertension.



Adrenergic antagonists inhibit the interaction of endogenous NE and epinephrine, and also sympathomimetic drugs, with adrenergic receptors at effector organs.

Alpha Blockers:

- Effects of blocking α_1 and α_2 adrenergic receptors
- Epinephrine Reversal

Alpha blockers block the vasopressor effects of epinephrine and norepinephrine.

Antagonism of α ARs:

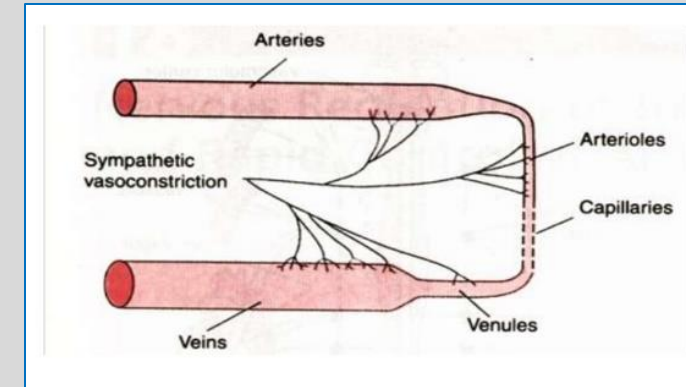
Selective and nonselective alpha blockers block α_1 receptor-mediated contraction of arterial, venous, and visceral smooth muscle.

Nonselective alpha blockers also block α_2 receptor-mediated presynaptic and postsynaptic actions.

α_2 actions are:

- suppression of sympathetic output
- increased vagal tone
- platelet aggregation
- inhibition of the release of NE and ACh from nerve endings
- regulation of metabolic effects (suppression of insulin secretion and inhibition of lipolysis)
- **contraction** of some arteries and veins by stimulation of post-synaptic α_2 ARs located on vascular smooth muscle.

(yes, vasoconstriction but it is less pronounced effect than α_1 AR- mediated contraction)

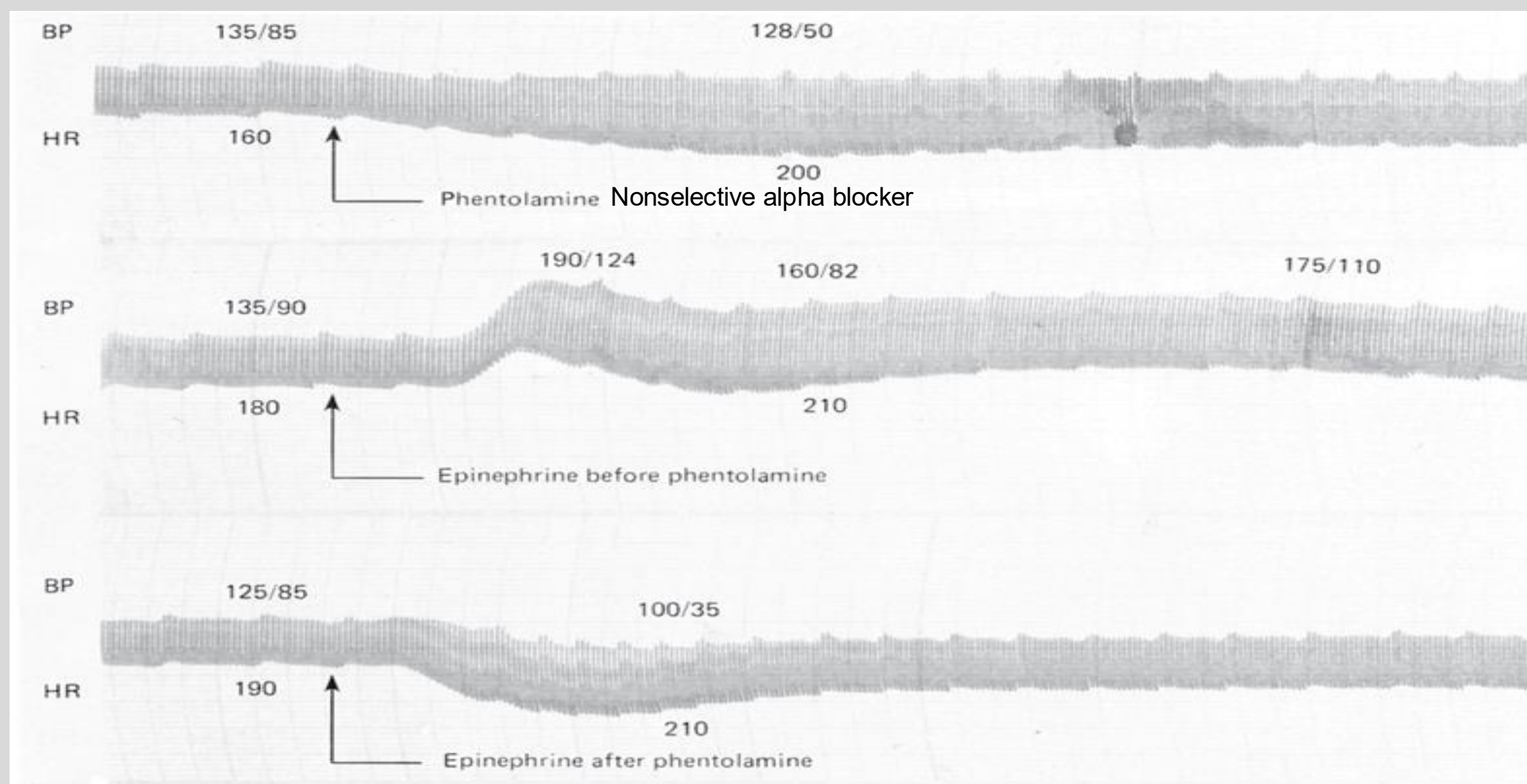


All of these α_2 -mediated actions are **BLOCKED**.

Alpha Blockers Class Effects

1. **Epinephrine reversal:** Vasopressor (“pressor”) response converted to depressor response
 - Alpha blockers’ antihypertensive use is based on this predictable effect.
2. **Cardiovascular effects:**
 - Blocking α_1 ARs $\Rightarrow$ direct relaxation of arteriolar and venous tone $\rightarrow$ $\downarrow$ peripheral vascular resistance $\rightarrow$ $\downarrow$ blood pressure
 - Therapeutic application: Chronic primary hypertension (add-on agents)
3. **Urinary effects:** Blockade of α_1 ARs $\rightarrow$ relaxation of the smooth muscle tone in the prostate and bladder neck $\rightarrow$ decreased resistance to the flow of urine
 - Therapeutic application: Relief of urinary symptoms due to benign prostatic hyperplasia (BPH)

What is the phenomenon described by these tracings?



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology*, 12th edition: www.accessmedicine.com

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Administration of an α blocker prior to admin of epinephrine converts EPI's pressor response into a depressor response. Inhibiting α AR agonist-induced vasoconstriction allows EPI's unopposed β_2 vasodilation $\rightarrow$ a reduction in blood pressure.

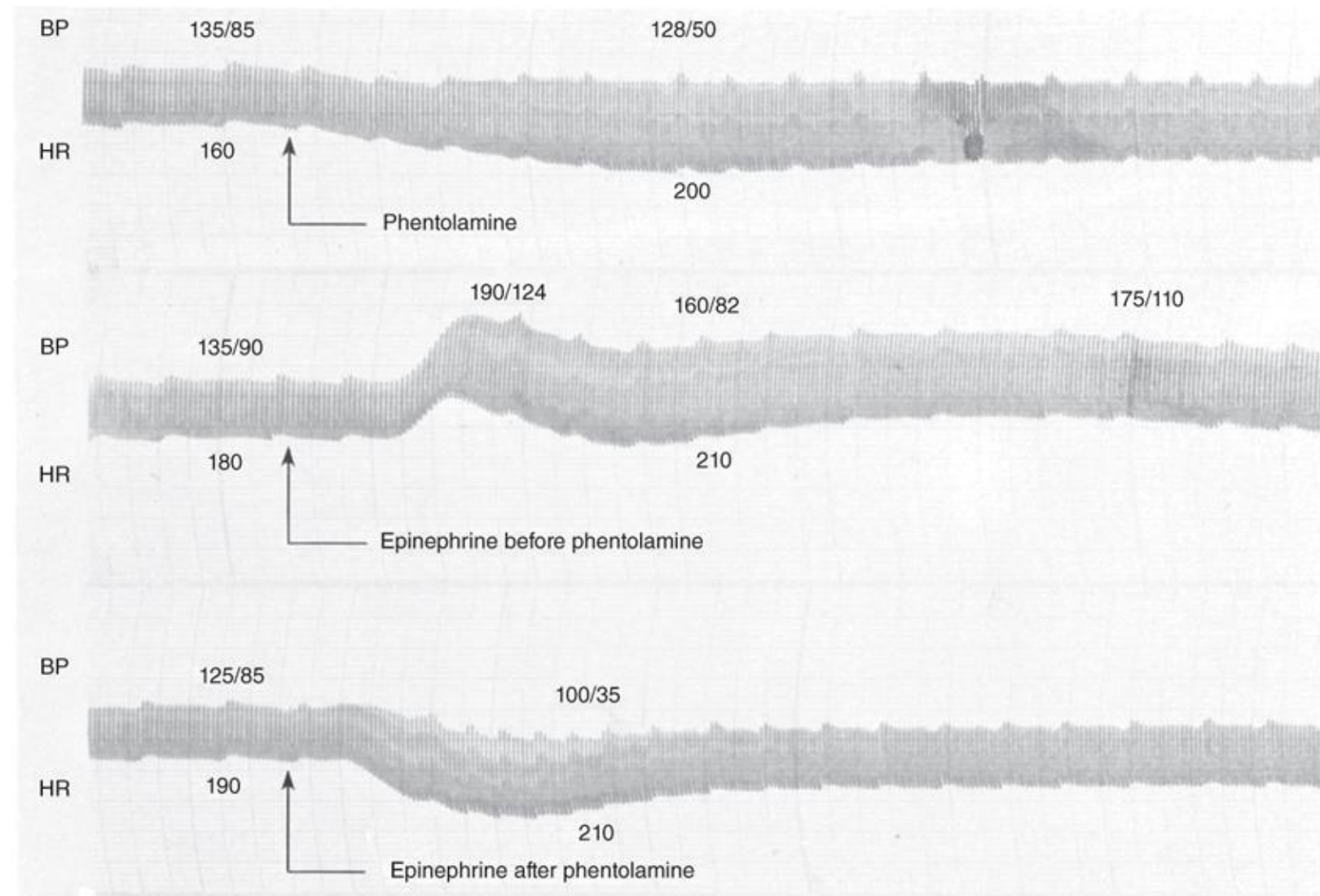
Epinephrine Reversal

Top: Effects of phentolamine, an α -receptor-blocking drug, on blood pressure in an anesthetized dog.

Epinephrine reversal is demonstrated by tracings showing the response to epinephrine before (middle) and after (bottom) phentolamine.

(Intravenous administration)

BP, blood pressure; HR, heart rate.

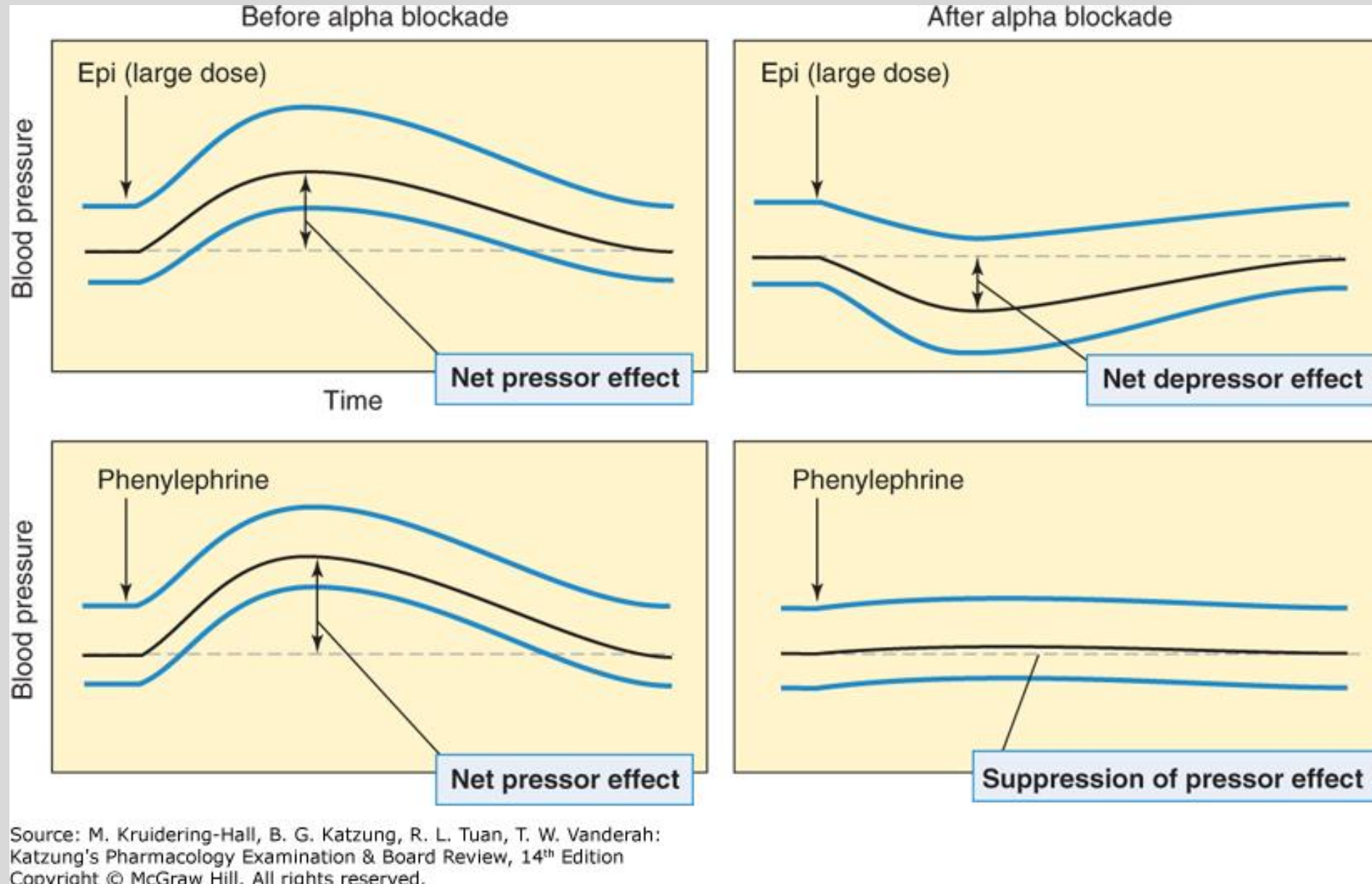


Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Compare the responses of EPI and phenylephrine to alpha blockade.

The effects of an α blocker, for example, **phentolamine**, on the blood pressure responses to **epinephrine** (epi) and **phenylephrine**.

- Top: The epinephrine response exhibits reversal of the mean blood pressure change from a net increase (the α response) to a net decrease (the β_2 response).
- Bottom: The response to phenylephrine is suppressed but not reversed, because phenylephrine lacks β action.



Non-Selective Alpha Blockers

*Phenoxybenzamine

*Phentolamine

Pharmacokinetics

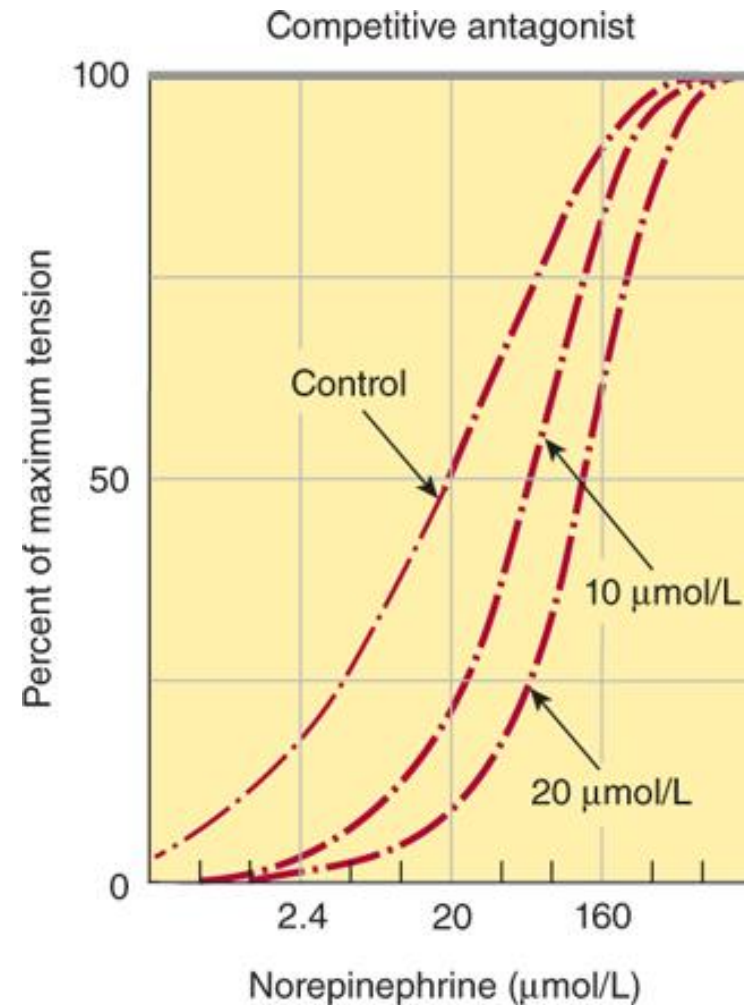
	Phenoxybenzamine	Phentolamine
A	Oral only 2x daily; Titrate dose gradually	IV, IM, submucosal infiltration (dental use to reverse local anesthetic)
D	May accumulate in adipose tissue	Not listed
M	Not known Bioavailability: 20-30%	Hepatic
E	Urine / bile	Urine
$t_{1/2}$	$t_{1/2\text{elimination}} \sim 24 \text{ h}$	IV ~ 20 minutes
	Onset: $< 2 \text{ h}$ Duration: 3-4 days (irreversible antagonist)	IV 1-2 minutes IM, Submucosal injection: 2-3h Duration IV 10-30 min; IM 30-45 min Peak submucosal 10-20 min
	PK values in patients treated for pheochromocytoma.	

Dose-response curves of norepinephrine in the presence of two different α -adrenoceptor–blocking drugs.

Left:

Tolazoline, a reversible blocker, shifted the curve to the right without decreasing the maximum response when present at concentrations of 10 and 20 $\mu\text{mol/L}$.

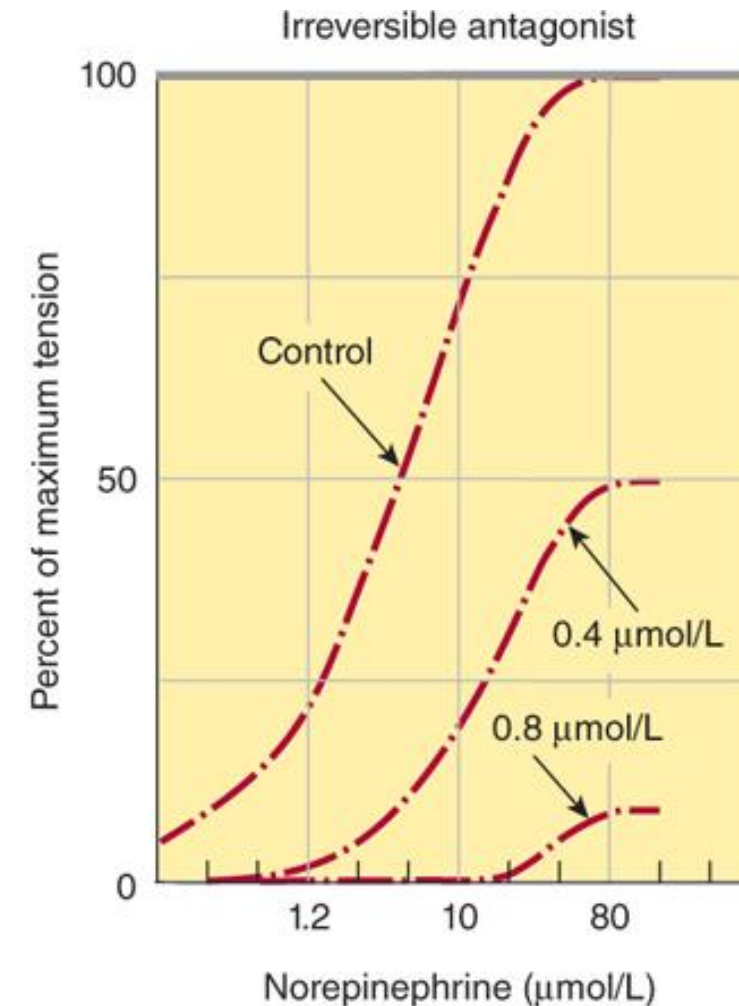
Phentolamine is a competitive, potent, nonselective α -adrenergic antagonist



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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The tension produced in isolated strips of cat spleen, a tissue rich in α receptors, was measured in response to graded doses of norepinephrine.

(Adapted with permission from Bickerton RK: The response of isolated strips of cat spleen to sympathomimetic drugs and their antagonists, J Pharmacol Exp Ther 1963 Oct;142:99-110.)



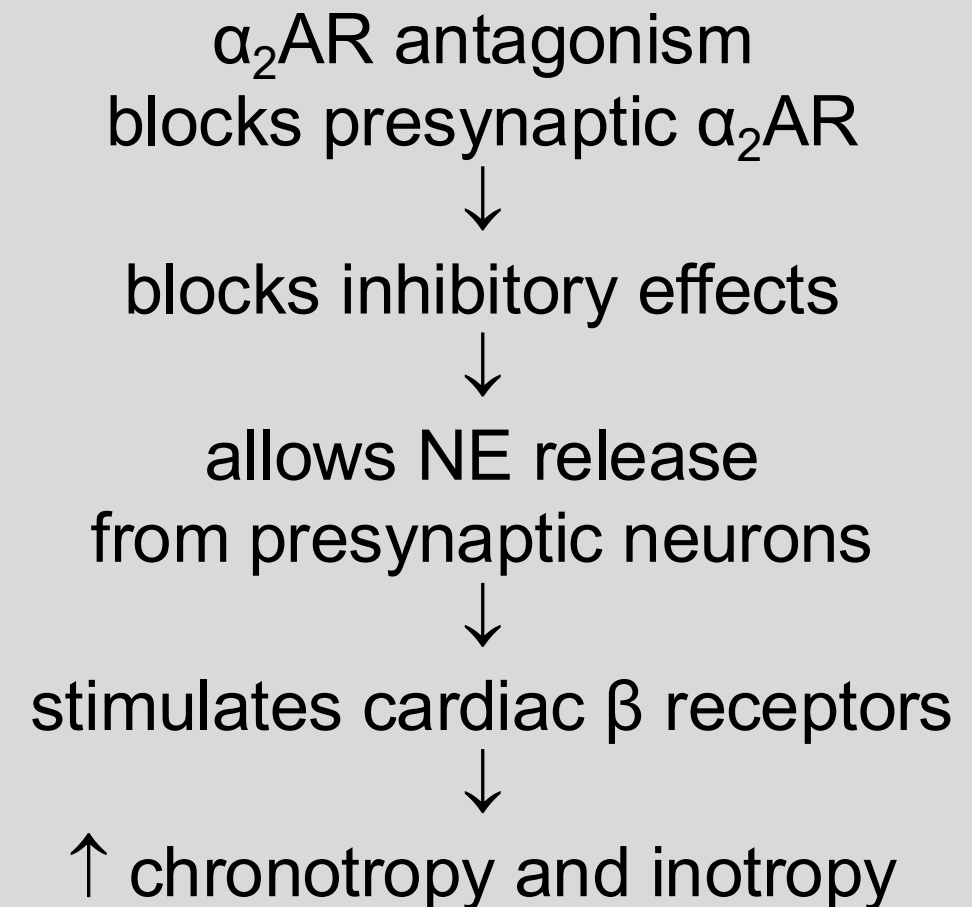
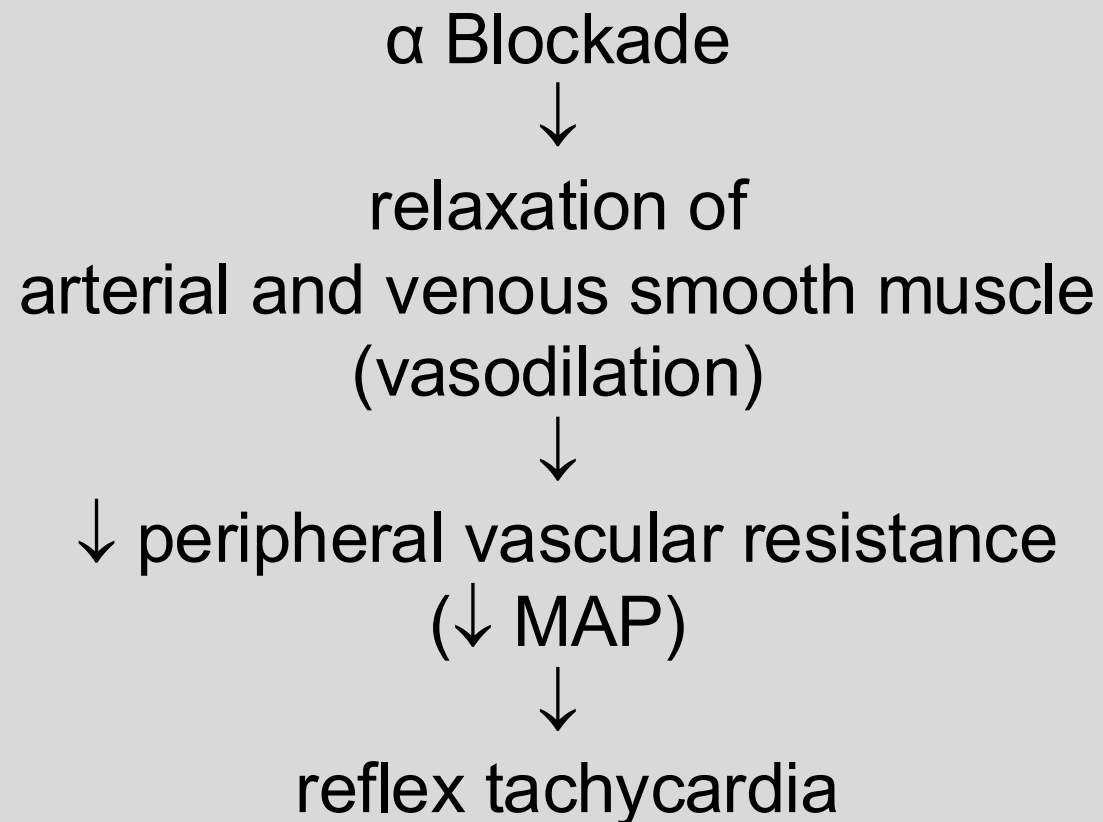
Right:

Dibenamine, an analog of phenoxybenzamine and irreversible in its action, reduced the maximum response attainable at both concentrations tested.

Phenoxybenzamine is an irreversible αAR antagonist (some selectivity for $\alpha_1\text{ARs}$)

Effects of nonselective alpha blockers

Cardiovascular actions and effect



Urinary retention:

Relaxes smooth muscle of prostate and bladder neck → relieves urinary retention

Clinical Applications of Alpha Blockers

Antagonism of EPI and NE at α ARs for:

Phenoxybenzamine

- Presurgical management of hypertension and sweating associated with pheochromocytoma
- Treatment of urinary symptoms associated with benign prostatic hypertrophy

Phentolamine

- Intraoperative hemodynamic instability: Rapid, short-term control of acute hypertension in patients undergoing surgery for pheochromocytoma
- Extravasation of a pressor agent (infiltration)
- Reversal of oral soft tissue (dental) local anesthetic containing epinephrine for local vasoconstriction (infiltration)

Alpha Blockers Class Adverse Effects (related to α AR block)

Vascular effects

1. Orthostatic hypotension:

- Blockade of α ARs prevents venoconstriction needed to maintain blood pressure in upright position.

2. Tachycardia: (nonselective agents)

- Inhibition of presynaptic α_2 ARs enhances release of NE, increasing heart rate
- $\downarrow$ MAP $\rightarrow$ central activation of sympathetic and inhibition of parasympathetic nervous systems $\rightarrow \uparrow$ heart rate

Nonvascular effects

- CNS: Sedation or depression
- Miosis
- Nasal stuffiness
- $\uparrow$ GI tract motility $\rightarrow$ abdominal pain, diarrhea
- Impaired/retrograde ejaculation

Alpha Blockers Drug Interactions

Effect

Examples

Drugs that increase blood pressure may diminish the effects of alpha blockers

Vasopressors ✓
Amphetamines ✓

Antihypertensive drugs from different classes may enhance the hypotensive effects of each other.
(may be beneficial)

Calcium channel blockers
ACE inhibitors, ARBs
Beta blockers ✓
Diuretics

Drugs with blood pressure lowering side effects may enhance the hypotensive effects of the alpha blockers

Phosphodiesterase-5 (PDE-5) inhibitors for erectile dysfunction
Antipsychotics
Antidepressants

Students, What you need to be able to do at this point in your learning is to explain the mechanisms of the drug interactions. You don't need to memorize the names of drugs you haven't learned about yet.

You have been introduced to the drug classes indicated with a ✓ and should know these.

Selective α AR Antagonists

α_1 -Selective		α_2 -Selective
Quinazolines for the treatment of HTN	Drugs for treatment of urinary symptoms of BPH	
*Prazosin • Terazosin • Doxazosin These alpha blockers are also used for patients with hypertension and concomitant urinary symptoms due to BPH.	*Tamsulosin (α_{1A} blocker) • Silodosin • Alfuzosin These drugs are NOT used in the treatment of hypertension.	*Yohimbine (botanical product)

Alpha-1 Blockers: Therapeutic Uses and Effects

Affinity	Affinity for α_1ARs is $\sim 1000\times > \alpha_2$ARs - Quinazolines have similar potencies at α_{1A}, α_{1B}, and α_{1D} subtypes - Tamsulosin has higher affinity for α_{1A}AR and α_{1D}AR
Hypertension add-on therapy, not first-line	Quinazoline alpha-1 blockers relax arterial and venous smooth muscle $\rightarrow \downarrow$ total peripheral resistance, $\downarrow$ preload, $\downarrow$ afterload. <i>Quinazolines have low tendency to increase cardiac output and rate.</i>
Benign prostatic hyperplasia (BPH)	α_{1A} blockers relax smooth muscle in bladder neck and prostate $\rightarrow$ improve urine flow and BPH urinary symptoms. (α_{1A} subtype may mediate prostatic smooth muscle contraction.) Tamsulosin has less pronounced effect than the quinazolines on standing blood pressure (may be mediated by α_{1B} subtype).

Alpha-1 Blockers Adverse Effects

First-dose effect: orthostatic hypotension, can be severe

- palpitations, tachycardia, dizziness, light-headedness, and syncope
- Advise patients to take the medication at bedtime.
- Orthostatic hypotension may occur for the first several doses.
- Caution also with the selective α_{1A} receptor blockers of BPH treatment.

Intraoperative floppy iris syndrome (IFIS) during cataract surgery

- α_1 AR inhibition causes relaxation of iris dilator muscle

Class nonvascular adverse effects:

Sedation or depression	↑GI tract motility → abdominal pain, diarrhea
Miosis	Impaired ejaculation
Nasal stuffiness	Tamsulosin: sulfonamide moiety → possible ↑risk of cross-reactivity in patients with h/o sulfonamide allergic

Yohimbine / Yohimbe bark Competitive α_2 AR antagonist

Yohimbine's effects are the opposite of clonidine's effects

1. Central effects (crosses blood-brain barrier): $\uparrow$ blood pressure and $\uparrow$ heart rate
2. Enhances motor activity and produces tremors
3. Sexual stimulant / Stimulating effect on mood

ADVERSE EFFECTS: related to presynaptic α_2 -adrenergic blockade:

tachycardia, hypertension

nausea, vomiting, anorexia, salivation (cholinergic effect)

antidiuretic action (stimulates ADH secretion)

tremors, diaphoresis

headache, agitation, anxiety, irritability

Questions

- How does the concept of epinephrine reversal explain the antihypertensive actions of alpha blockers?
- What is the reason (and mechanism) for advising patients to take the first dose of a selective alpha blocker at bedtime?
- Selective inhibition of which adrenergic receptor type - α_1 , α_2 , β_1 , β_2 , or β_3 - leads to tachycardia, hypertension, nausea, vomiting, anorexia, salivation, tremors, diaphoresis, headache, agitation, anxiety, irritability? What drug?

β -Adrenergic Receptor Antagonists

1. Classical nonselective β blockers (*Propranolol)
2. β_1 -Selective β blockers (*Metoprolol)
3. Nonselective β blockers with additional actions
4. β_1 -Selective β blockers with additional actions

Additional actions include α_1 ARs block, β_2 AR stimulation, enhanced nitric oxide (NO) production, block Ca^{2+} entry, open K^+ channels, or possess antioxidant properties; these additional properties may offer therapeutic advantages.

Pharmacodynamics of Beta Blockers (BB)

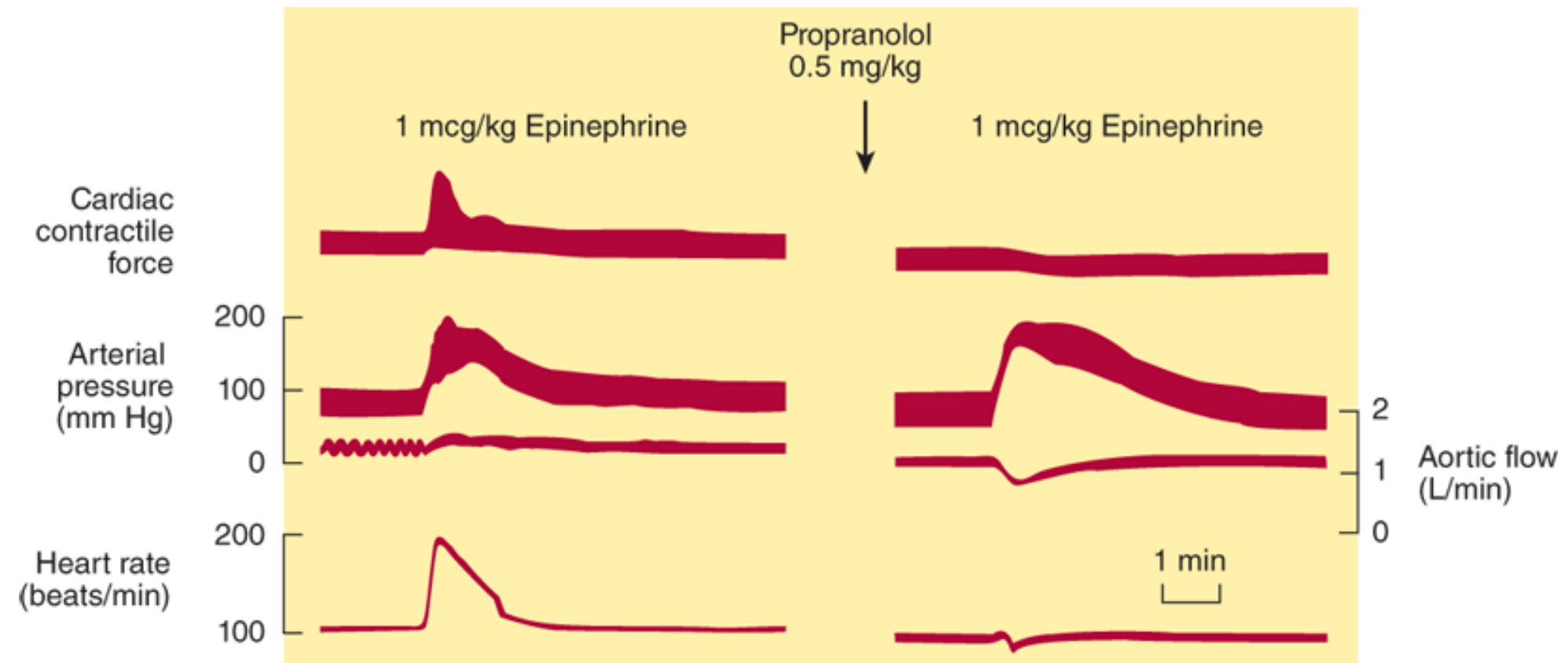
The various beta blockers may have two or three but not all of the actions.

*Competitive antagonists	prevent EPI / NE from binding active site
* β_1 / β_2 Receptor blocking actions	nonselective vs. β_1 -selective drugs
Intrinsic sympathomimetic activity	partial agonist / antagonist
Membrane stabilizing activity	prevent action potentials from being propagated across the membrane
Inverse agonist activity	same effects as the competitive antagonist
α_1 -Receptor antagonism	vasodilation effect
Nitric oxide production	NO is a potent vasodilator
Block L-type voltage-gated Ca^{2+} channels	arteriolar vasodilation
Antioxidant actions	may contribute to clinical benefits in chronic heart failure

BB Cardiovascular Effects: In the presence of a β AR-blocking agent, EPI no longer augments the force of contraction nor increases cardiac rate.

Blood pressure is still elevated by EPI because vasoconstriction is not blocked.

The effect in an anesthetized dog of the injection of epinephrine before and after propranolol. In the presence of a β -receptor–blocking agent, epinephrine no longer augments the force of contraction (measured by a strain gauge attached to the ventricular wall) nor increases cardiac rate. Blood pressure is still elevated by epinephrine because vasoconstriction is not blocked.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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(Reproduced with permission from Shanks RG: The pharmacology of beta sympathetic blockade, Am J Cardiol 1966 Sep;18(3):308-316.)

Beta Blockers Effects on Cardiac Conduction

(Figures are on the next slide.)

1. Slow sinus rate (negative chronotropic effect)
2. ↓ the spontaneous rate of depolarization of ectopic pacemakers (↓automaticity)
3. Reduce contractility (negative inotropic effect)
4. Slow conduction in the atria and atrioventricular (AV) node → ↑ PR interval
5. ↑ the functional refractory period in the AV node

The takeaway:

Beta blockers slow heart rate, conduction, and force of contraction.

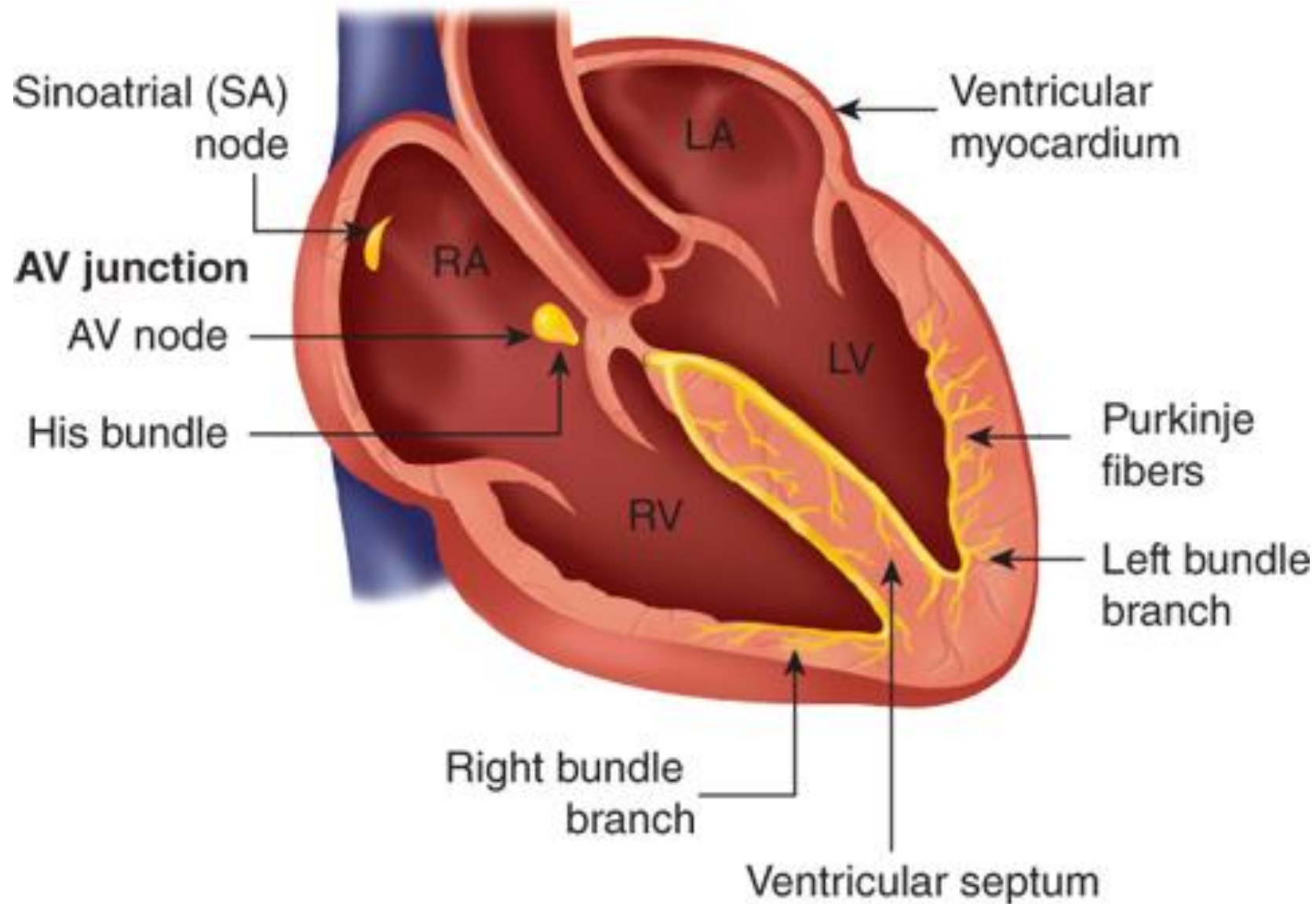
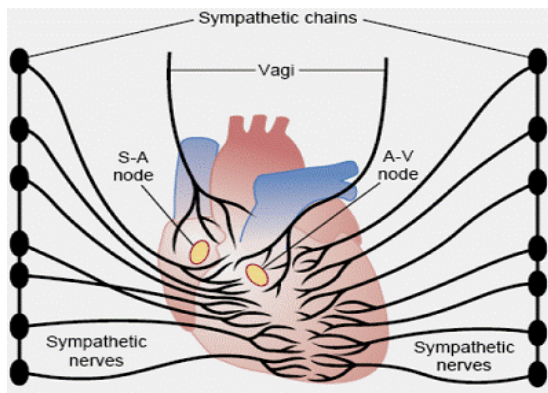
These actions reduce cardiac output, which decreases blood pressure.

Short-term use of nonselective beta blocker (propranolol):

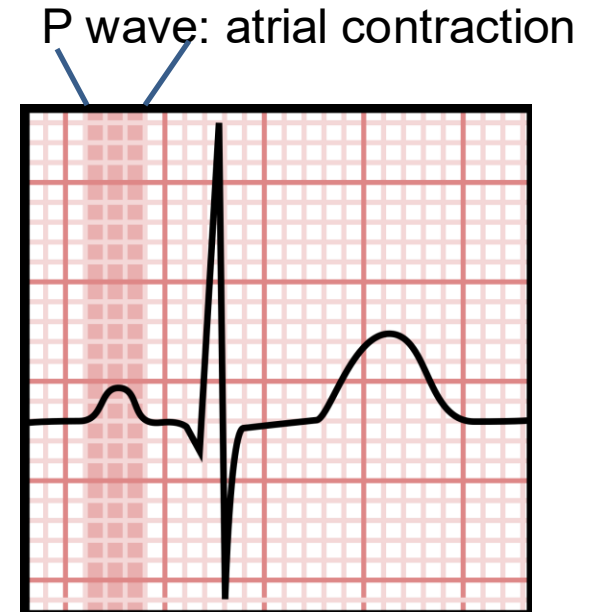
↓ cardiac output → ↓ BP → reflex ↑ peripheral resistance due to block of β_2 ARs → unopposed α -mediated vasoconstriction

Long-term use of nonselective beta blocker → total peripheral resistance returns to initial value or decreases in patients with hypertension

Renin-angiotensin system: suppression of renin release prevents synthesis of the hormone angiotensin II, a powerful vasoconstrictor.



Source: Sylvia C. McKean, John J. Ross, Daniel D. Dressler, Danielle B. Scheurer: Principles and Practice of Hospital Medicine, Second Edition, www.accessmedicine.com
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Cardiac conduction system

Therapeutic uses of Beta Blockers

Hypertension	Decrease cardiac output and blood pressure, decrease renin release
Angina pectoris	Decrease oxygen consumption and the work of the heart
Acute myocardial infarction	Decrease sympathetic input to the heart and improves morbidity and mortality when given early during AMI
Congestive heart failure	Decrease sympathetic input to the heart and improves morbidity and mortality in CHF with reduced ejection fraction
Arrhythmias	Treatment of supraventricular and ventricular arrhythmias
Hyperthyroidism	Reduce tachycardia (due to upregulation of cardiac β receptors), palpitations, tremulousness, and anxiety
Esophageal varices	Decrease portal hypertension and the incidence of bleeding and mortality in patients with hepatic cirrhosis
Glaucoma	Reduces aqueous humor formation by ciliary body
Migraine prophylaxis	Reduce frequency and intensity of migraine
Performance anxiety	Reduce the somatic manifestations of anxiety

Potential adverse effects (most are minor)

Bradycardia

Bronchospasm (asthma, COPD) due to β_2 block
 β_2 block may potentiate or mask hypoglycemia
in diabetic patients prone to hypoglycemia

Inhibits reflex hepatic production of glucose

Masks warning signs (eg, rapid heart beat,
tremors, sweating, anxiousness)

Lipid abnormalities: modest effect on
triglycerides and HDL-C

Insomnia; fatigue; depression, sexual
dysfunction (based on poor studies and subject
of debate)

 **Abrupt discontinuation of beta blocker after long-term treatment can cause severe angina, myocardial infarction or ventricular arrhythmia, and may increase the risk of sudden death.**

- Supersensitivity mediated by β AR adaptive upregulation
- *Taper the dose of the beta blocker over several weeks.*

Precautions / Drug Interactions

Contraindications

- Asthma
- Hypersensitivity to beta blocker

Pregnancy

- Cross placenta
- Other antihypertensive agents are preferred

Exception: Labetalol is used for hypertension management in the pregnant patient.

Drug interactions

- Other antihypertensive agents enhance blood pressure lowering effect (often beneficial)
- Some calcium channel blockers increase risk of heart block
- CYP inducers / inhibitors
- NSAIDs increase Na^+ and water retention – attenuates antihypertensive effect
- Other drugs with effects on BP

Thought Questions

- How do the beta blockers' inhibitory effects on cardiac cells reduce blood pressure?
- What is the mechanistic thread that connects all the therapeutic uses of the beta blockers?
- How does the mechanism of action describe the mechanisms of adverse effects associated with beta blockers?

Pharmacokinetic Properties of Propranolol and Metoprolol

*Propranolol (nonselective)

*Metoprolol (β 1-selective)

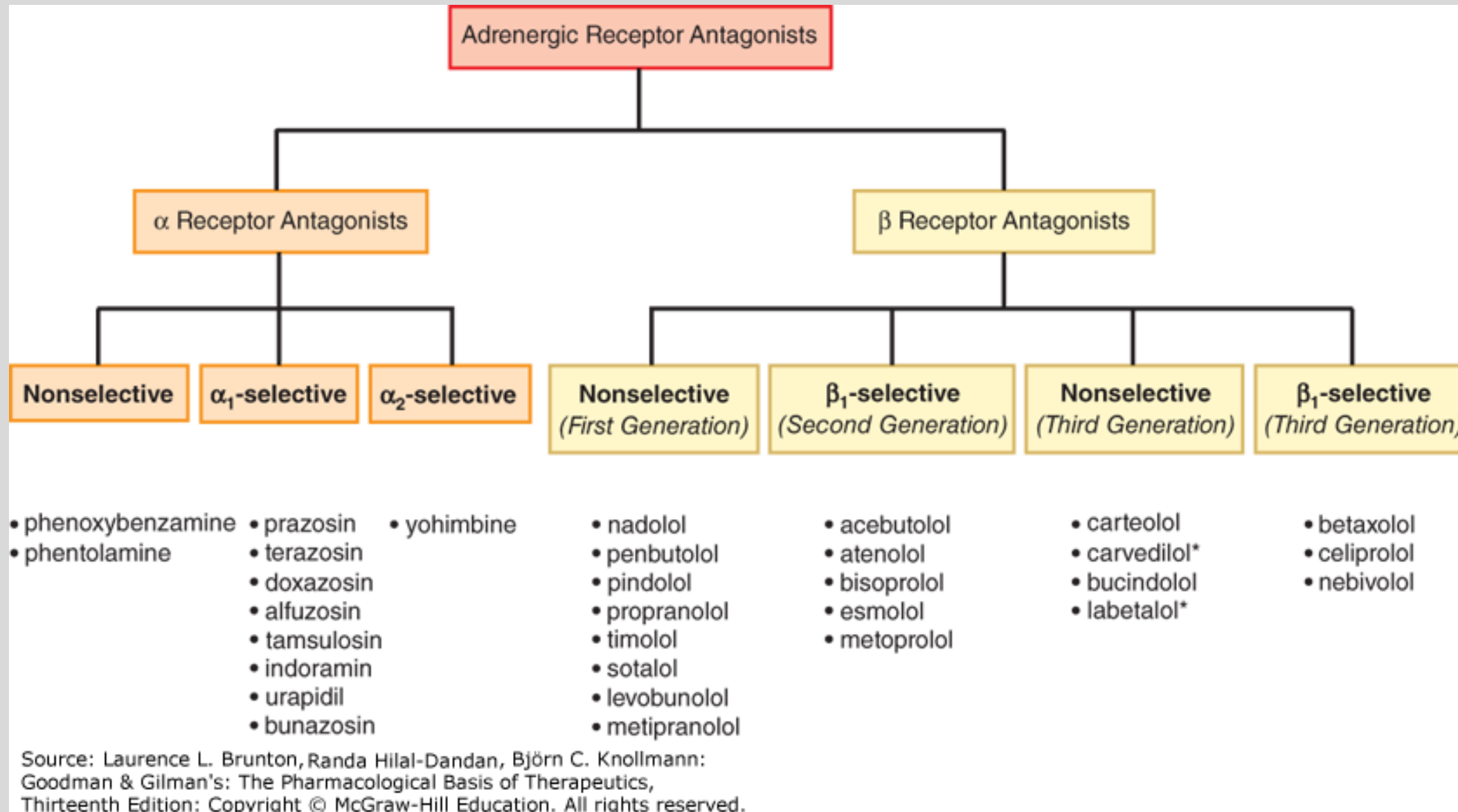
A	Oral Completely absorption from GI tract	Oral, IV Complete absorption from GI tract
D	Protein binding~ 90% Crosses BBB	Protein binding ~10% Crosses BBB
M	CYP1A2, 3A4, others Active metabolite	CYP2D6
F	~25%; ~50% if taken protein-rich foods Immediate release tab: take on empty stomach	~50%
E	Urine; metabolites	Urine, metabolites
t $\frac{1}{2}$	3-6 hours (adults, immediate release form)	~4 hour ~8 hours in CYP2D6 poor metabolizers

Values in adults, immediate release formulations; extended release forms are available

BBB: blood-brain barrier

Lexicomp monographs

Summary of Adrenergic Antagonists



Adrenergic antagonists inhibit the interaction of endogenous NE and epinephrine, and also sympathomimetic drugs, with adrenergic receptors at effector organs.

- The α AR antagonists (alpha blockers) relax vascular smooth muscle and decrease peripheral vascular resistance. They reduce the pressor effects of α agonists and can convert epinephrine's vasopressor effects to a depressor response. Epinephrine reversal is a predictable effect of alpha blockade.
- The β AR antagonists (beta blockers) reduce heart rate, conduction velocity, force of contraction, and cardiac output.
- Both alpha and beta blockers reduce blood pressure.
- The nonselective alpha blockers include phenoxybenzamine (an irreversible inhibitor) and phentolamine (a competitive blocker). Phenoxybenzamine is primarily used to manage presurgical hypertensive and sweating episodes caused by pheochromocytoma. Phentolamine is for rapid, short-term use in intraoperative severe hypertension in pheochromocytoma patients, reversal of extravasation of pressor agents, and reversal of oral soft tissue local anesthesia. They cause marked tachycardia and marked orthostatic hypotension.
- The alpha-1 blockers are used as add-on agents for the management of primary hypertension and they are used for the treatment of urinary symptoms due to benign prostatic hypertrophy (BPH). They cause orthostatic hypotension and syncope (first-dose effect), intraoperative floppy iris syndrome, miosis, nasal stuffiness, and impaired/retrograde ejaculation.

- β_1 -Receptors are primarily found in the heart and kidneys. In the heart, they control heart rate and contractility. Long-term use of a beta blocker reduces blood pressure and myocardial oxygen demand. β_2 -Receptors are primarily found in the lungs, skeletal muscle vasculature, and liver. Blocking β_2 -receptors results in increased airway resistance and metabolic changes that can result in hypoglycemia and hyperkalemia.
- Beta blockers have prominent effects on the heart, decreasing heart rate (chronotropy) and force of contraction (inotropy), and they suppress renin release from specialized cells in the kidney. CNS effects likely contribute to their blood pressure lowering effects.
- Frequently used beta blockers are the nonselective agents that block both β_1 and β_2 receptors, cardioselective (β_1 only) agents, and drugs with combined α and β blocking actions, such as labetalol.
- Beta blockers are useful in the treatment of a number of cardiovascular diseases, such as arrhythmia, effort angina (ischemic heart disease), acute myocardial infarction, congestive heart failure, and as add-on agents in the treatment of hypertension. Beta blockers are also used in the treatment of tachycardia associated with hyperthyroidism, esophageal varices, migraine prophylaxis, and symptoms of performance anxiety.
- Adverse effects include bronchoconstriction (β_2 blockade), bradycardia, decreased cardiac output, hypoglycemic effects and masking of hypoglycemia, and lipid abnormalities. Depression, fatigue, and sexual dysfunction have been reported.

Thought Questions

- How does the concept of epinephrine reversal explain the antihypertensive actions of alpha blockers?
- Blocking vascular α ARs prevents EPI's vasoconstrictive actions, leading to vasodilation. EPI's β 2-mediated vasodilation is unopposed, contributing to the antihypertensive actions of alpha blockers.
- What is the reason (and mechanism) for advising patients to take the first dose of a selective alpha blocker at bedtime?
- The first dose (or first few doses) may cause orthostatic hypotension because, when rising from a lying down to upright position, reflex vasoconstriction is blocked.
- Selective inhibition of which adrenergic receptor type - α 1, α 2, β 1, β 2, or β 3 - leads to tachycardia, hypertension, nausea, vomiting, anorexia, salivation, tremors, diaphoresis, headache, agitation, anxiety, irritability? What drug?

Thought Questions

- How do the beta blockers' inhibitory effects on cardiac cells reduce blood pressure?
- Beta blockers decrease cardiac output by decreasing heart rate, slowing conduction through the AV node, and decreasing the force of contraction.
- What is the mechanistic thread that connects all the therapeutic uses of the beta blockers?
- Beta blockers inhibit the stimulating effects of the sympathetic nervous system on β ARs.
- How does the mechanism of action describe the mechanisms of adverse effects associated with beta blockers?
- Bradycardia (all BBs). Blocking the β_2 ARs can exacerbate asthma/COPD and potentiate or mask symptoms of hypoglycemia.

Resources

- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e 2023, Chapter 14: Adrenergic Agonists and Antagonists
- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024: Chapter 10: Adrenoceptor Antagonist Drugs
- UpToDate Lexidrug monographs

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